Final Project Proposal

CAI 5720 | Initial proposal
Due November 9, 2026, at 11:59 PM ET

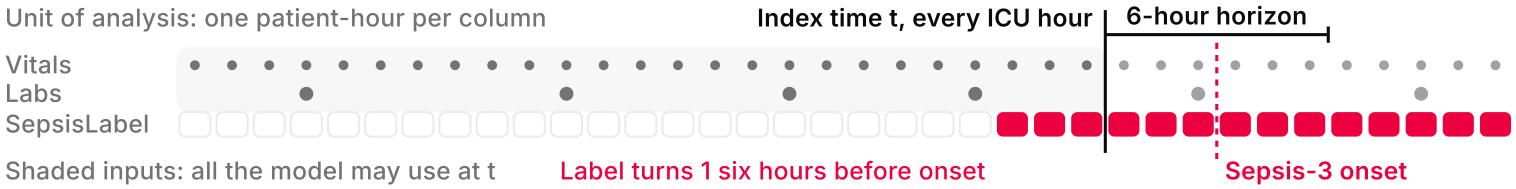
Working project title: **Does an early sepsis warning hold up at a hospital it has never seen?**

Canvas group and members: Final Project 6 (Second Site); Karissa Kocourek, Daniel Escalante

Research question and clinical significance

Using only data charted up to a given ICU hour, can a model flag adult ICU patients who will meet Sepsis-3 criteria within the next 6 hours, and does it stay accurate and calibrated at a hospital it was never trained on?

It serves bedside nurses and rapid response teams deciding whether to start a sepsis workup. Earlier sepsis treatment improves survival, yet Wong et al., JAMA Intern Med 2021, found a widely deployed commercial model performed far worse in external validation than its developers reported, a gap we want to measure and explain.



Proposed dataset

PhysioNet/Computing in Cardiology Challenge 2019, Early Prediction of Sepsis from Clinical Data, version 1.0.0, at physionet.org/content/challenge-2019/1.0.0, open access under CC BY 4.0, no credentialing, described in Reyna et al., Crit Care Med 2020. Over a million hourly rows hold 8 vital signs, 26 labs and 6 demographic or admission variables, with heavy lab missingness. Target: hourly SepsisLabel; positive class: sepsis.

Beth Israel Deaconess	20,336 patients, 8.8% sepsis	Emory	20,000 patients, 5.7% sepsis
Development: patient-level cross-validation		External test: untouched until everything is locked	

Planned methods

Split and leakage control. Each patient's hours stay in one fold, and Emory is opened only after the model, predictor set and threshold are locked. Imputation, scaling and any recalibration are fit within development folds only; patient ID and anything recorded after the index time are excluded. Predictors are the latest values, short-term trends, and indicators of whether each lab has been measured so far.

Models. Training-prevalence reference | SIRS-style rule | Logistic regression | Gradient boosted trees

Tuning. Each learned model gets a small cross-validated search; logistic regression is the interpretable one.
Evaluation on Emory. AUROC, AUPRC as average precision, ROC and precision-recall curves, calibration curve, Brier score, and a confusion matrix with sensitivity, specificity and PPV at a threshold set in development data from a fixed alert budget, plus alerts per patient-day and patient-level bootstrap confidence intervals.
Interpretation. Standardized logistic coefficients, permutation importance, two contrasting held-out patients including a missed case, errors by age group and sex with counts, and a sensitivity analysis without the measurement indicators to test whether the model reads physiology or clinician ordering habits.

Expected outcome

We expect both learned models to lose discrimination and especially calibration at Emory, with part of their signal reflecting when labs were ordered rather than lab values. We will produce a reproducible Colab notebook and report on the performance drop at a new hospital and the alerts a unit would face. Limits: charted-data labels, retrospective design, two academic centers, no race or treatment data. **Still tentative:** the alert budget level, dropping post-onset hours, subsampling hours per patient, and also training on Emory to test on Beth Israel.

Smaller fallback if sepsis is too large: Early Stage Diabetes Risk Prediction, 520 records, detecting recorded diabetes from symptoms; or Chronic Kidney Disease, 400 records, though its labs may mirror the label.